

## Bivariate data with technology



1

a  $y = -0.29x + 0.67$

b  $y = 0.12x - 1.75$

2

a  
i  $-0.47$

ii Weak negative linear correlation

b  
i  $-0.84$

ii Strong negative linear correlation

c  
i  $0.47$

ii Weak positive linear correlation

d  
i  $-0.91$

ii Strong negative linear correlation

e  
i  $-0.59$

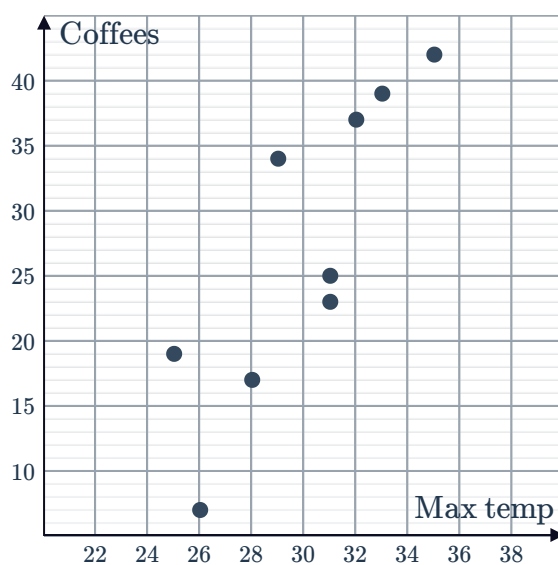
ii Moderate negative linear correlation

f  
i  $0.19$

ii Weak positive linear correlation

3

a



b  $r = 0.84$

c As the temperature increases, the sales tend to increase.

4

a  $-0.85$

b Strong negative linear relationship

c  $y = -0.1x + 30.3$

5



- a 0.60
- b Moderate positive linear relationship
- c  $y = 4.2x - 34.5$

6

- a  $y = -1694x + 18\,407$
- b 18 407
- c \$18 407
- d -1694
- e The gradient of the line indicates that age and price have a negative correlation.
- f The price decreases by \$1694.

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## Predictions

7

- a  $r = -0.908$
- b Strong negative linear relationship
- c  $y = -0.06x + 6.37$
- d 6.07 kg
- e Reliable due to interpolation and a strong correlation.

8

- a  $r = 0.997$
- b Strong positive linear relationship
- c  $y = 0.85x + 1.28$
- d 52.28%
- e Reliable due to interpolation and a strong correlation.

9

- a  $V = 14.9P + 30.3$
- b 30.3
- c 30.3 words
- d Yes, as children learn words in other ways.
- e 14.9
- f For each additional page that a child reads each day, the child's vocabulary will increase by 14.9 words.

10



- a  $P = -0.2M + 99.6$
- b 99.6
- c 99.6%
- d Yes, it is reasonable because they will have more time to study.
- e  $-0.2$
- f For each additional minute that a student uses SnappyChatty, their test mark will decrease by 0.2%.

11

- a  $r = 0.79$
- b Strong positive linear relationship
- c  $y = 0.025x - 550.108$
- d \$1449.89
- e \$80 000 because the other two are examples of extrapolation.

12

- a  $r = 0.86$
- b Strong positive linear relationship
- c  $y = 0.89x + 2.78$
- d  $25.03^{\circ}\text{C}$
- e  $32^{\circ}\text{C}$ , because the other two values are examples of extrapolation.

13

- a  $r = -0.91$
- b Strong negative linear relationship
- c  $y = -6.6x + 80.1$
- d 47.1
- e Reliable due to interpolation and a strong correlation.

14

- a 0.72
- b Strong positive linear relationship
- c  $A = 1.61E - 16.63$
- d 39.72%
- e C

15



- a 0.86
- b Strong positive linear relationship
- c  $y = 0.98x - 7.94$
- d \$50.86
- e Reliable due to interpolation and a strong correlation.

16

- a 0.67
- b Moderate positive linear relationship
- c  $y = 0.21x - 2.09$
- d 15 computers
- e Very unreliable due to extrapolation and a moderate to weak correlation.

17

- a  $r = -0.20$
- b  $(1, 1.28)$
- c  $r = 0.79$
- d Strong positive linear relationship
- e  $y = 0.05x + 0.23$
- f 1.23
- g Despite a moderate correlation, unreliable due to extrapolation.